F70LB Life Insurance Mathematics B

Heriot-Watt University
Chapter 3 Excel modelling

2025-2026

When new graduates start work with us, the best ones quickly learn that doing a calculation well isn't just a matter of getting an answer out. A good calculation is easy to read and gives me confidence that it is correct. I want the graduate to follow clear, consistent spreadsheet standards and to include evidence of self-checking.

The calculation should be easy to adapt and reuse, and hard to accidentally break. The most talented graduates also appreciate that a model is just a model. It contains judgments, assumptions and approximations. Much of the path to becoming an actuary is about developing judgment.

An actuary, 2024

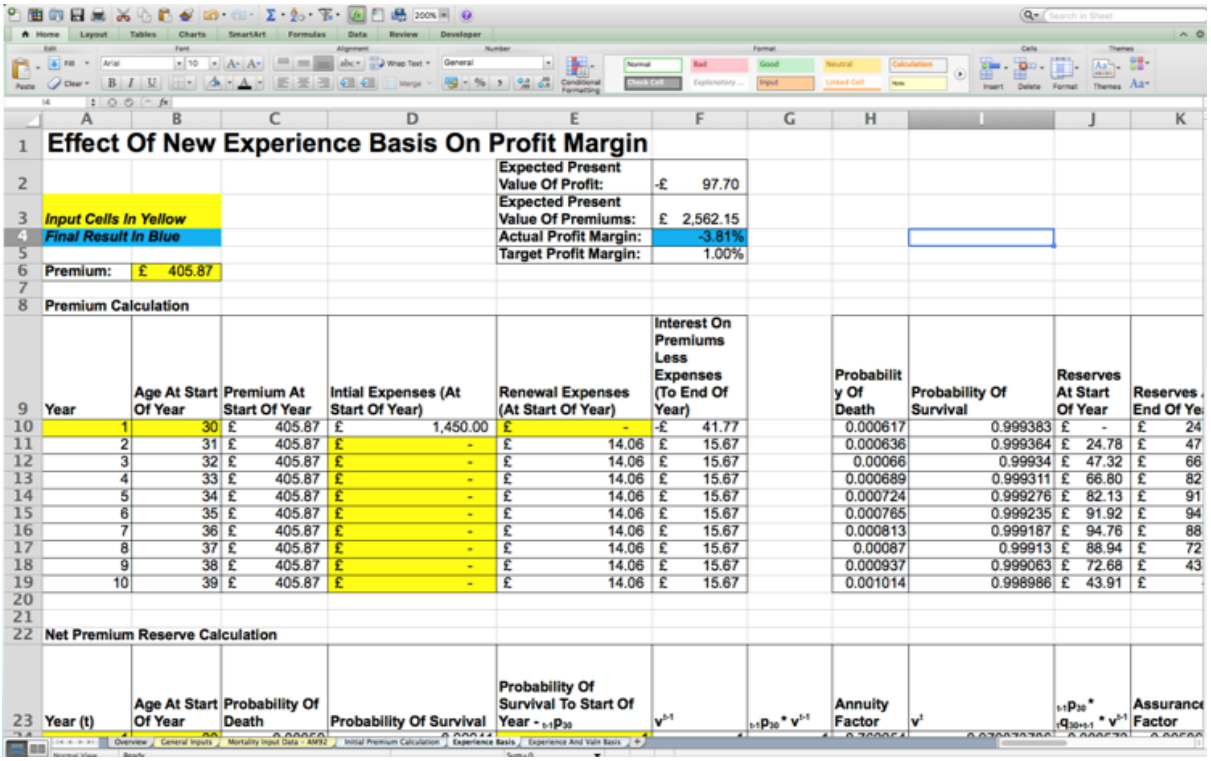
3.1 Introduction

In the workplace, you need your colleagues to have confidence in your calculation work. As the quotation above shows, clear and consistent formatting and documentation of your calculations is necessary and highly valued. In the early stages of your career, your time will be worth less that your boss's, so you want to save them work by presenting calculations that are easy to verify. You should be able to answer the question "How do you know this is the right answer?"

Actuaries make frequent use of Excel, Microsoft's spreadsheet software. It is used to record information, carry out calculations and perform analyses on the calculations. The illustration below is an example of what an Excel worksheet might look like. A single Excel spreadsheet will typically have multiple worksheets.

We spend much of this chapter discussing how to set up an Excel spreadsheet professionally. We also list out the main Excel tools and techniques you will be expected to know. Make sure you review this thoroughly so that you are prepared for your Excel assessment.

Finally, we will look at how to implement some important actuarial calculations in Excel. You will have the opportunity to practice this in labs.



Effect Of New Experience Basis On Profit Margin

Expected Present Value Of Profit:	-£ 97.70
Expected Present Value Of Premiums:	£ 2,562.15
Actual Profit Margin:	-3.81%
Target Profit Margin:	1.00%

Input Cells In Yellow
Final Result In Blue

Premium: £ 405.87

Premium Calculation

Year	Age At Start Of Year	Premium At Start Of Year	Intial Expenses (At Start Of Year)	Renewal Expenses (At Start Of Year)	Interest On Premiums Less Expenses (To End Of Year)	Probability Of Death	Probability Of Survival	Reserves At Start Of Year	Reserves End Of Ye
1	30	£ 405.87	£ 1,450.00	£ -	-£ 41.77	0.000617	0.999383	£ -	£ 24
2	31	£ 405.87	£ -	£ 14.06	£ 15.67	0.000636	0.999364	£ 24.78	£ 47
3	32	£ 405.87	£ -	£ 14.06	£ 15.67	0.00066	0.99934	£ 47.32	£ 66
4	33	£ 405.87	£ -	£ 14.06	£ 15.67	0.000689	0.999311	£ 66.80	£ 82
5	34	£ 405.87	£ -	£ 14.06	£ 15.67	0.000724	0.999276	£ 82.13	£ 91
6	35	£ 405.87	£ -	£ 14.06	£ 15.67	0.000765	0.999235	£ 91.92	£ 94
7	36	£ 405.87	£ -	£ 14.06	£ 15.67	0.000813	0.999187	£ 94.76	£ 88
8	37	£ 405.87	£ -	£ 14.06	£ 15.67	0.00087	0.99913	£ 88.94	£ 72
9	38	£ 405.87	£ -	£ 14.06	£ 15.67	0.000937	0.999063	£ 72.68	£ 43
10	39	£ 405.87	£ -	£ 14.06	£ 15.67	0.001014	0.998986	£ 43.91	£

Net Premium Reserve Calculation

Year (t)	Age At Start Of Year	Probability Of Death	Probability Of Survival	Probability Of Survival To Start Of Year - $_{t-1}p_{30}$	v^{t-1}	$_{t-1}p_{30} * v^{t-1}$	Annuity Factor	v^t	$_{t-1}p_{30} * q_{30+t-1} * v^{t-1}$	Assurance Factor

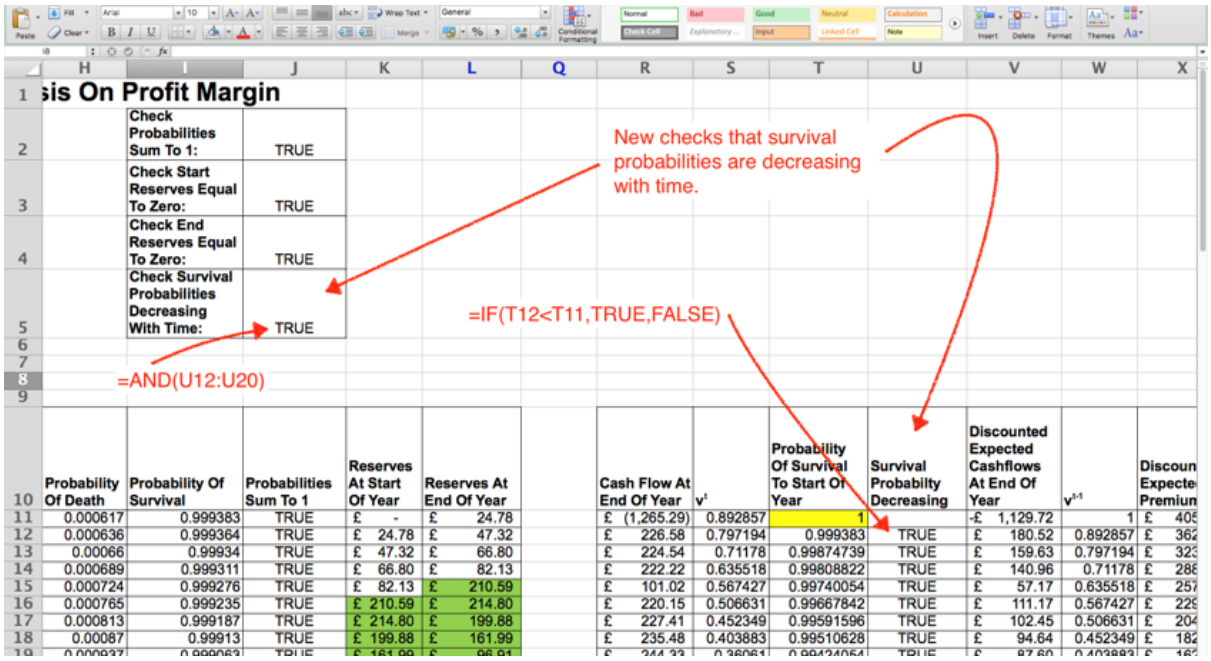
3.2 Best practice for setting up a spreadsheet

All spreadsheets should be easy to follow and understand. Any documentation that goes with the spreadsheet should also be easy to understand. The following guidelines should be considered when setting up a spreadsheet:

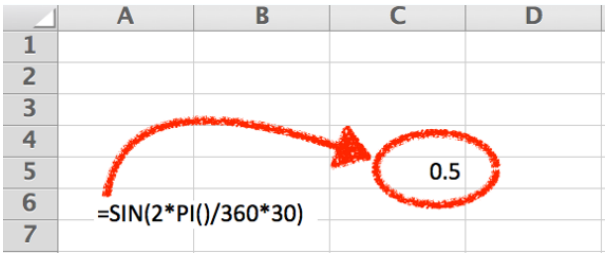
1. Each worksheet should be read like a book - from left to right and top to bottom.
2. Each column and row should therefore have a single purpose.
3. Set up your spreadsheet with one column for each data type. The column should start with a heading and then have individual data entries in each row.
4. The worksheet names ('tabs') should have short names with the explanation of the spreadsheet purpose included in the the audit trail.
5. Label your columns/rows clearly.
6. Use blank columns/rows to avoid the spreadsheet becoming too cluttered.
7. Choose appropriate number of decimals to display. Showing too many decimal places can make the numbers hard to read which can also lead to errors.
8. Avoid hard-coding numbers in your formula. Use variables and formulae instead where possible.
9. Make your spreadsheet flexible so that additional calculations can be included with ease and so that additional data points can be added.
10. Use different cell shading colours to distinguish between cells that are inputs, cells that are parameters/assumptions and cells that are formulae.
11. Model parameters are usually included on one worksheet where possible.
12. Use a different cell shading colour to highlight where a manual adjustment has been made to the data.

13. Break up complicated calculations (don't try to do lots of calculations in one cell.

14. Incorporate checks into your calculations. For example, incorporate an automatic check that a column of data containing ages has no negative values or that survival probabilities are decreasing over time (like in the illustration below).



Example:



This is a spreadsheet for calculating sine of 30 degrees.
Question - What is wrong with this spreadsheet?

Answer - There is/are no:

- Title
- Author
- Version number
- Date of last update
- Instructions on how to use the spreadsheet
- Clear inputs
- Clear outputs

Hard-coding is also used which means it cannot be easily changed.
Here is a better version of the above spreadsheet:

	A	B	C	D	E	F
1	**Spreadsheet for Calculating Sine of an Angle in Degrees**					
2	by Jane Gormley					
3	Version 1					
4	Last Updated:	01/02/2024				
5						
6	***Input cells are shown in yellow***					
7	***Output cells are shown in blue***					
8						
9	**Angle**	30	degrees	=SIN(2*PI()/360*B9)		
10						
11	**Sine of the Angle**	0.5				
12						

3.3 Useful Excel functions

The following tools/functions are often useful in performing Excel calculations. You should ensure that you are comfortable with everything in the following lists.

3.3.1 Summary statistics:

- AVERAGE - calculates the average (arithmetic mean)
- MAX - gives the highest value
- MIN - gives the lowest value
- COUNT- counts the number of cells that contain numbers
- SUM- calculates the total

3.3.2 Logical operators

- IF - evaluates a condition and returns one value if condition is true and another value if condition is false.
- AND- outputs 'TRUE' if all of the conditions are true and outputs 'FALSE' if any of the conditions are false.
- OR -outputs 'TRUE' if one of the conditions are true and 'FALSE' if all of the conditions are false.

3.3.3 Number formatting

- ROUND - rounds a number to a specified nuber of digits.
- ABS - outputs the absolute value of a number.

3.3.4 Analysis functions

- COUNTIF- counts the number of cells within a range that meet a specific condition.
- PRODUCT- multiplies all numbers in a range of cells and outputs the result.
- SUMPRODUCT- multiples corresponding components in specified arrays and reutrns the sum of these products. It is useful when calculating weighted sums and weighted averages.
- SUMIF- adds the cells specified by a given criteria.
- VLOOKUP- searches for a value in the leftmost column of a table and outputs a value in the same row from a specified column.
- HLOOKUP- searches for a value in the top row of a table and outputs a value in the same column from a specified row.
- XLOOKUP- searches a range and outputs an item corresponding to the first match found. This function is generally more flexible and robust than VLOOKUP or HLOOKUP which are older functions.

3.3.5 Other useful tools

You would also be expected to know how to do the following in Excel:

- Use goal seek
- Sort data
- Apply a filter
- Prepare graphs/charts

3.4 Some important examples in Excel

This final section of the chapter before we conclude sets out how to implement some of the more common or important actuarial calculations in Excel. We start with numerical methods for integration, which you may have seen before, and move on to annuities, assurances and expectation of life.

3.4.1 Numerical methods

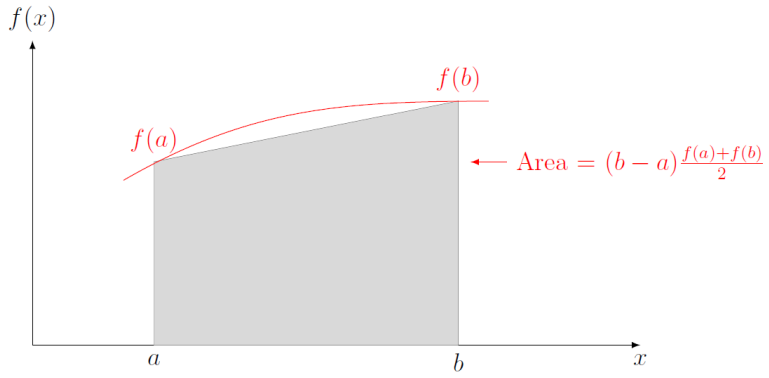
In this section we discuss two methods for numerical integration, Trapezium Rule and Simpson's Rule. These are useful for calculating numerical answers in Excel to functions such as $\bar{a}_{x:\overline{n}|} = \int_{t=0}^{t=n} e^{-\delta t}{}_t p_x \, dt$

Trapezium Rule

The Trapezium Rule is

$$\int_a^b f(x)\,dx \approx (b-a)(f(a)+f(b))/2$$

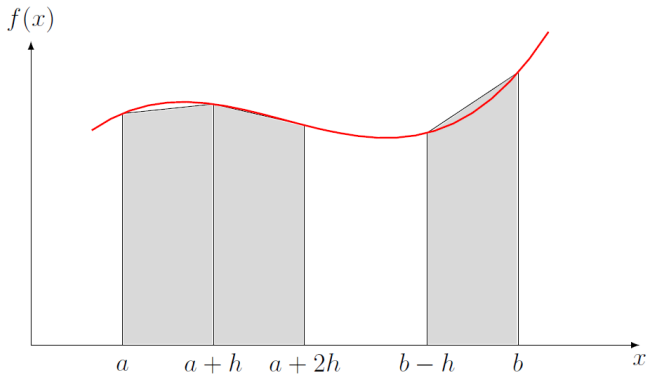
You can visualise this as follows:



So if $f(x)$ were a linear function, the Trapezium Rule would give us exactly the correct answer. For non-linear functions, we can get a more accurate approximation by splitting the area into many trapeziums, instead of just one. We call this the Repeated Trapezium Rule.

To use the Repeated Trapezium Rule, choose a positive integer, n, preferably large, and let $h = (b-a)/n$. Then:

$$\begin{aligned}
\int_a^b f(x)\,dx &= \int_a^{a+h} f(x)\,dx + \int_{a+h}^{a+2h} f(x)\,dx + \ldots + \int_{b-h}^{b} f(x)\,dx \\
&\approx h(f(a) + f(a+h) + f(a+h) + f(a+2h) + \ldots + f(b-h) + f(b))/2 \\
&= h\left(\frac{f(a)}{2} + f(a+h) + f(a+2h) + \ldots + f(b-h) + \frac{f(b)}{2}\right)
\end{aligned}$$



We can implement the formula

$$\int_a^b f(x)\,dx = h\left(\frac{f(a)}{2} + f(a+h) + f(a+2h) + \ldots + f(b-h) + \frac{f(b)}{2}\right)$$

in Excel to estimate

$$\bar{a}_{40:\overline{1}|} = \int_{t=0}^{t=1} e^{-\delta t}\,{}_tp_{40}\,dt$$

using a step size $h = 0.1$ as follows:

Calculation of EPV of a continuous annuity using Trapezium Rule

Parameters	
delta	0.07
x	40
stepsize h	0.1

Results

EPV annuity at time zero

0.965

Formula for EPV annuity is
*=SUM(G14:G24)*StepSize*

time	age	(t)p(x)	Weight	v^t	Weighted integrand annuity
0	40	1.00000	0.5	1.00	0.50000
0.1	40.1	0.99992	1	0.99	=D15*E15*F15
0.2	40.2	0.99985	1	0.99	0.98595
0.3	40.3	0.99977	1	0.98	0.97900
0.4	40.4	0.99970	1	0.97	0.97209
0.5	40.5	0.99962	1	0.97	0.96524
0.6	40.6	0.99954	1	0.96	0.95843
0.7	40.7	0.99946	1	0.95	0.95167
0.8	40.8	0.99938	1	0.95	0.94495
0.9	40.9	0.99930	1	0.94	0.93829
1	41	0.99922	0.5	0.93	0.46583

The column "Weighted integrand annuity" is $w_t f(t)$ where $f(t) = e^{-\delta t}\,{}_tp_{40}$ and the weights w_t in column E are 0.5 at the end points and 1 at all intermediate points. We add these up and multiply through by the stepsize $h = 0.1$ in the results cell $F7$ to get our final answer.

There are various different ways we could reasonably set this out in Excel. However, note some elements of good practice:

- Parameters and results are clearly separated and identified. Parameters are not hard coded.
- Our calculations are set out from top to bottom and left to right
- We use a few columns to avoid long formulas that would be harder to read. (We could certainly accomplish this calculation in fewer cells, but that might not be better.)

- Columns are labelled with clear headings, but the aim is clarity not necessarily "beauty". (You should not usually waste time on anything that does not have a practical purpose.) For example, it's fine to have written (t)p(x) not ${}_tp_x$.
- All the cells are colour-coded in a meaningful way. Many organisations have a "house style" which helps anyone in the organisation open someone else's spreadsheet and quickly interpret it.

The specific colour key used in this spreadsheet was:

Colour key
Parameters
Data/values
Calculations
Discontinuous formulas
Value looked up from data
Results

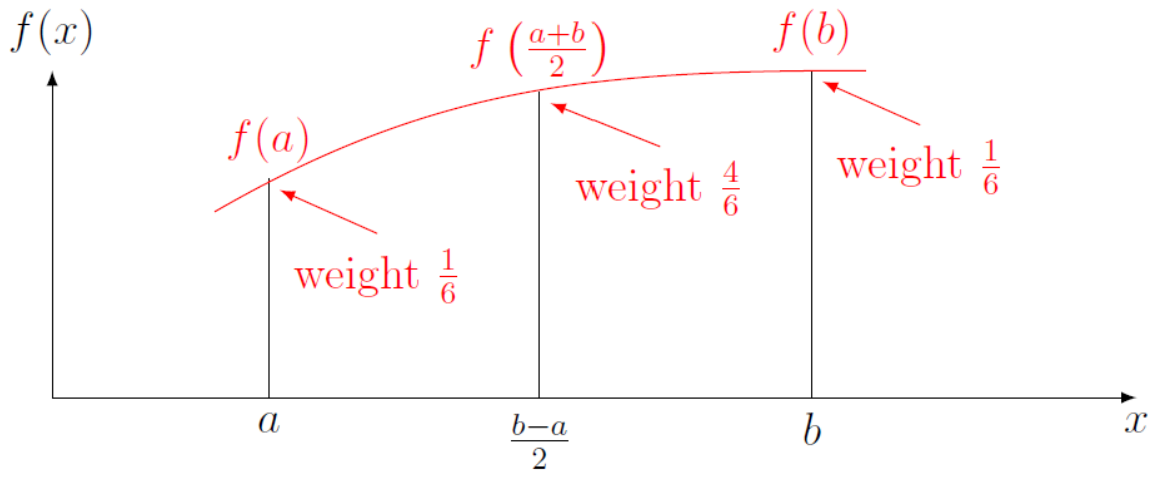
(Not all parts of the key were relevant.) Note that the weights were inputted as values. It would be possible to parameterise this as a formula instead.

Simpson's Rule

Another rule for numerical integration is Simpson's Rule. It approximates an integral by fitting a quadratic function to the end-points and mid-point of the range over which the function is integrated. Suppose f is a real-valued function and $a < b$ are constants. Then Simpson's Rule is

$$\int_a^b f(x)dx \approx \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

Simpson's Rule gives an exact answer if f is a polynomial of degree 3 or less.



To get a more precise answer when f is not a polynomial of degree 3 or less, partition the interval $[a, b]$ into sub-intervals and apply Simpson's Rule over each sub-interval. For example, fix a positive integer n and choose the partition so that each sub-interval is of length $(b-a)/n$. We call this the 'Repeated Simpson's Rule with n steps'.

Setting the length of each sub-interval to $h := (b-a)/n$, the integral to be evaluated can be split as follows.

$$\begin{aligned}\int_a^b f(x)\,dx &= \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + \cdots + \int_{b-h}^{b} f(x)dx \\ &\approx \frac{h}{6}(f(a) + 4f(a+h/2) + 2f(a+h) + 4f(a+3h/2) + \ldots + \\ &\qquad 2f(b-h) + 4f(b-h/2) + f(b))\end{aligned}$$

Applying Simpson's Rule to each of the integrals on the right-hand side has given us the Repeated Simpson's Rule with n steps.

Notice that this is really just the same as Repeated Trapezium Rule, but with different weights. It can be implemented in Excel in the same way, just changing the weights appropriately. The weights for Simpson's Rule are $1:4:2:4:2:\cdots:4:2:4:1$ (multiplied through by $\frac{h}{6}$)

Simpson's Rule works exactly for the integrals of $f(x) = x^3$, $f(x) = x^2$ and $f(x) = x$. It is suitable for estimating other integrals.

3.4.2 Annuities, assurances and expectation of life

In this final subsection before concluding the chapter, we look at calculating annuities, assurances and life expectancies in Excel. We start with the continuous case, then look at payments in discrete time.

Continuous-time payments and complete expectation of life

You have already seen in the previous section how to estimate the expected present value of a continuous annuity in Excel using the formula $\bar{a}_{x:\overline{n}|} = \int_0^n e^{-\delta t}\,{}_tp_x\,dt$ with the Repeated Trapezium Rule or Repeated Simpson's Rule.

You can calculate a value for a term insurance paid immediately on death in exactly the same way using $\bar{A}^1_{x:\overline{n}|} = \int_0^n e^{-\delta t}\,{}_tp_x\mu_{x+t}\,dt$. Just include μ_{x+t} in the integrand $f(t)$.

For whole-life annuities and assurances, simply extend your calculation to the final age of the life table you are using where the probability of survival is assumed to be zero. If, for example, this is age 120, then in Excel (or any other software) you can use $\bar{a}_x = \bar{a}_{x:\overline{120-x}|}$ and similarly for assurances.

The complete expectation of life is mathematically the same as a continuous annuity at the same age with the interest rate set to zero, so if you have an annuity calculation set up already, it is possible to use that. Alternatively, use $\mathring{e} = \int_0^{\omega-x} {}_tp_x\,dt$

Finally, an alternative approach is to calculate the discrete-time equivalents (see below) and use standard approximations which you know from last semester, such as:

$$\bar{a}_x \approx a_x + 0.5$$

$$\mathring{e}_x \approx e_x + 0.5$$

$$\bar{A}_x \approx (1+i)^{0.5} A_x$$

Discrete-time payments: Annuities in advance

There are broadly two ways to calculate an annuity or an assurance in Excel where the payments occur in discrete time (ie, at the end of the time period or at the start, not continuously or immediately). You can either use a sum or a recursive relationship and will arrive at the same answer. Let's start with the example of an annuity paid in advance. The other applications are similar and will be discussed at the end of the section.

For annuities paid in advance, you can use (for integers $n > 0$):

$$\ddot{a}_{x:\overline{n}|} = \sum_{k=1}^{n} v^{(k-1)}\,{}_{k-1}p_x$$

Or proceed recursively using:

$$\ddot{a}_{x:\overline{n+1}|} = \ddot{a}_{x:\overline{n}|} + v^n {}_np_x$$

taking $\ddot{a}_{x:\overline{0}|} = 0$ by definition.
We can implement this in Excel as follows:

	A	B	C	D	E	F
1	**Calculation of the EPV of a 10-year annuity paid in advance to a life aged 40**					
2						
3	**Parameters**			**Results**		
4				EPV a due(40:10) calculated as a sum		
5	interest rate	0.05		8.075	=SUM(E13:E23)	
6				EPV a due(40:10) calculated recursively		
7				8.075	=F23	
8				Check A-B=0 (ie, same answer)		
9				0.000		
10						
11						
12	time	age	(t)p(x)	v^t	(t-1)p(x) * v^(t-1)	EPV of adue(x:t)
13	0	40	1.00000	1.00	Not required	0
14	1	41	0.99922	0.95	1.000	1.000
15	2	42	0.99837	0.91	0.952	1.952
16	3	43	0.99744	0.86	0.906	=F15+C15*D15
17	4	44	0.99642	0.82	0.862	3.719
18	5	45	0.99531	0.78	0.820	4.539
19	6	46	0.99411	0.75	0.780	5.318
20	7	47	0.99280	0.71	0.742	6.060
21	8	48	0.99138	0.68	0.706	6.766
22	9	49	0.98984	0.64	0.671	7.437
23	10	50	0.98818	0.61	0.638	8.075

Column E is the summand $v^{(k-1)}{}_{k-1}p_x$ for each k and cell $D5$ shows the result of adding these to get

$$\ddot{a}_{x:\overline{n}|} = \sum_{k=1}^{n} v^{(k-1)}{}_{k-1}p_x = 8.075$$

Column F is the alternative recursive method. It starts from $\ddot{a}_{x:\overline{0}|} = 0$ in cell $F13$ and then uses

$$\ddot{a}_{x:\overline{k+1}|} = \ddot{a}_{x:\overline{k}|} + v^k {}_kp_x$$

at each the rows below.

We see at the bottom of the column (cell $F23$), and repeated in the Results section at the top (cell $D7$), that the recursive method has also given the result

$$\ddot{a}_{x:\overline{n}|} = 8.075$$

Note that:

- We have discontinuous formulas in cells $E13$ and $F13$, ie the formulas in the rows below those cells are different. It is important to highlight discontinuous formulas to stop the next user (or you at a later date) from accidentally breaking the spreadsheet.
- The Results box includes a check (in grey) that the two methods give us the same answer. Doing a calculation in two different ways is sometimes an appropriate way to check your calculations (or someone else's). However, it is often not possible or proportionate. Approximate checks are often more appropriate.

Discrete-time payments and curtate expectation of life

The calculation for annuities paid at the end of the year is similar to annuities in advance. For annuities paid at different intervals, you can break the calculation down into one step (row) per interval, eg monthly steps. Alternatively, it may be reasonable to do the annual calculation, then apply Woolhouse's approximation to the final result. For monthly or weekly payments, it is common in practice to use the expected present value of a continuous annuity, since this is sufficiently accurate for most purposes. Of course, no payment is truly continuous or instant!

The curtate expectation of life is in effect a special case of the EPV of an annuity paid annually in arrears with interest rate zero, and can be calculated in the same way (or by setting the interest rate parameter to zero).

For assurances paid at the end of each year, you can use:

$$A^1_{x:\overline{n}|} = \sum_{k=1}^{n} v^k {}_{k-1}p_x q_{x+k-1}$$

Or proceed recursively using:

$$A^1_{x:\overline{n+1}|} = A^1_{x:\overline{n}|} + v^{(n+1)} {}_np_x q_{x+n}$$

with $A^1_{x:\overline{0}|} = 0$

These can be set out in Excel in a similar way to annuities in advance shown above.

For an endowment assurance, you can proceed in a similar way in Excel to a term insurance, don't forget to include the maturity payment in addition to the term insurance element:

$$A_{x:\overline{n}|} = A^1_{x:\overline{n}|} + A_{x:\overline{n}|}{}^1$$

Optional exercise: Another way to calculate the EPV of an n-year term insurance or endowment assurance in Excel is to calculate $A^1_{x+t:\overline{n-t}|}$ (or $A_{x+t:\overline{n-t}|}$) recursively on each row starting from a terminal condition $A^1_{x+n:\overline{0}|} = 0$ (or $A_{x+n:\overline{0}|} = 1$). This is quite a neat method because only the terminal condition needs to be different! (The recursive formula is the same.) Can you think how to set this up in Excel? (HINT: If not, then come back to this after you've covered recursive policy values in Excel. It's the same!)

3.5 Conclusion

Your takeaways from this chapter should include:

- The way that you structure and present your Excel calculations is important. It affects the likelihood of error, ease of checking and whether your work can safely be reused. It is essential for working with others (and even your future self if you open your work in 6 months' time!). You may be assessed in this.
- Make sure that you are comfortable with all the Excel tools/functions listed in this chapter. These may be required in assessments.
- How to implement some key numerical techniques and actuarial functions in Excel. (You will practice this in labs.)